6DATA007W – Final Project Report

Project Title

Student: first and last name (student number)

Supervisor: first and last name

This report is submitted in partial fulfillment of the requirements for the

BSc (Hons) Data Science and Analytics

at the University of Westminster

School of Computer Science & Engineering
University of Westminster

Date

Document Scope

The purpose of this document is to describe and reflect on the processes that took place for developing the Final Project, discuss any ethical issues associated with it, describe the methodology that was adopted to develop the project.

All chapter word counts in this document are approximate and are not intended to be prescriptive.

All sections in orange must be deleted and removed before submitting the report.
Any text in red should be replaced by appropriate information.

Declaration

This report has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Word Count: The final word count
Student Name: Your full name
Date of Submission: Date

This is an important section!
Add the updated word count (do not count words in the Acknowledgments, Table of Contents, Table of Figures, Table of Tables, References, Bibliography and Appendix).

Use of Generative AI

In this assessment, I used [list tools] to [state usage].
You may refer to the document entitled "*Guidance for students on the use of Generative Artificial Intelligence*".

Abstract

500 words

Summarise here the problem statement and the project aim(s). Provide a brief description of the methodology followed, main results, your conclusions and observations.

Acknowledgements

Thank those you who helped you build your project and supported you during its development, if you wish to here.

Table of contents

Before submitting, update the table of contents above!
Word should automatically populate and update pages, if you follow the instructions available here.

List of Figures

Provide a list of figures, linking figure numbers to page numbers. If you can, hyperlink the page numbers/figures. *If you have no figures in your document, please remove this page.*

List of Tables

Provide a list of tables (if any), linking table numbers to page numbers. If you can, hyperlink the page numbers/tables. If you have no tables in your document, please remove this page.

1. Introduction

Introduce the project, include the problem statement, project aim(s) and objectives.

1.1 Problem statement

500 words

Give some background on the problem you intend to address and the need for model(s)/algorithm(s)/software/application. Use references to support your statements, when possible, and illustrations, diagrams and figures, if needed.

1.2 Aims and Objectives

300 words

The aim(s) describe, in a few sentences, the overarching purpose(s)/intention(s) of the model(s)/algorithm(s)/software/application. What is the point of developing the model(s)/algorithm(s)/software/application, what you wish to achieve?

Objectives describe with some details the individual steps you will take to fulfil the project aim(s).

2. Background

Include a literature survey in the research topic, discuss existing relevant or similar studies and/or applications to yours with an emphasis on methods and techniques used, and review tools that are used to tackle projects similar to yours.

2.1 Literature review

800 words

Describe initial results of a literature survey on a selected research topic/application area related to your project subject. Use relevant books, published research articles as well as Internet content for the purpose. Make use of in-text references to indicate your sources.

Table 1: Add caption explaining the table

	methods	Dataset	Key insights
Article 1			
Article 2			

2.2 Review of methods and applications

800 words

Describe your background research on relevant existing studies and methods/algorithms/techniques, their advantages and disadvantages. This section should also include a review of software/applications/systems if you intend to develop one. Use illustrations, diagrams, screenshots, graphs for the purpose.

A comparison table may be provided to distinguish the key characteristics of methods/algorithms/techniques relevant to your project.

You may also produce a Table of Features, comparing the main features of the software/applications/systems and the one you developed, if relevant.

2.3 Review of tools

800 words

Describe results of a survey on relevant tools that can be used to develop models and/or applications such as the one you built for your project, e.g., programming languages and environments, libraries. List their advantages and disadvantages. Use illustrations, diagrams, screenshots for the purpose.

This section is about all relevant tools to your project including selected/used tools and their alternatives.

3. Legal, Ethical, Sustainability and other considerations

800 words

Consider any legal, ethical, sustainability, social, professional and security considerations associated with your research, the data you are collecting/analysing and/or the software/application you are building.

4. Methodology

1500 Words

Describe fully the approach you followed to conduct the project. This may include:

- List and present the key tasks to conduct, clarifying what algorithms and techniques are used for each task. Examples of tasks: data collection, data pre-processing, model development…
- If you intend to develop a software/application/system, present an appropriate software development methodology for your project and discuss why it is suited for your project. Examples of methodologies: Agile (SCRUM, DSDM…), Waterfall…

4.1 Data

Provide a presentation of the data used, e.g., data source, data collection period, number of observations, number and types of features/variables …summary statistics…

4.2 Methods

Present algorithms/methods/techniques used in this project.

If you intend to develop a software/application/system, you may have to use/create ERD, flowcharting, storyboarding, prototyping…

Discuss in some detail issues relating to (if relevant):

- User Interface
- Infrastructure
- Functionality
- Algorithm development
- Content creation
- Other

5. Tools

500 Words

Describe the tools (programming environments & languages, libraries) **you used** for the development of your model(s)/algorithm(s)/ software/ application/ system. Justify your choices.

6. Model development

2500 words

This section consists in presenting model development.

You may add and comment on key extracts of code which is even more important in case you are developing a software/application/system. The code would be presented by use case or with reference to requirements. You can include pseudo-code or/and code essential snippets.

Explain implementation of main code for key functions, indicate any novel code clearly and code that is adopted/adapted and the original sources.

In case you are developing a software/application/system, you have to present/discuss test coverage and test methodology. Present and discuss what was tested. Discuss how the output was tested and why.

7. Results, analysis and discussion

1500 words

Analyse and discuss key results. Evaluation and comparison of (analytical) models using performance metrics as appropriate.

Figure 1: Add caption explaining the figure

8. Conclusions and reflections

1000 words

Provide critical reflections on ALL aspects of the project lifecycle. Include conclusions on the resulting research/model(s)/findings/software/application, reflect on strengths and limitations, State existing skills development, discuss the acquisition of any new knowledge and skills for building your project and consider any further work.

9. References

Include a list of items (books, papers, websites, etc.) cited in your text. Use Cite them Right (either Harvard or IEEE) for the purpose.

10. Bibliography

Include here a list of general reading items (books, papers, websites, etc.). List the items in alphabetical order, using Cite them right Harvard style to describe them.

Appendix A

Provide additional material, if appropriate, in separate appendices (e.g. Appendix A, Appendix B…or Appendix A1, Appendix A2…).

Use one Appendix to provide a link to an on-line video demo of the project and a link to your project GitHub page.

For **video recording**, we recommend the following:

- Set your display resolution to a standard size (1920×1080 is ideal).
- Close unnecessary windows/tabs to keep the recording clean and professional.
- Use full-screen mode when demonstrating code, dashboard, software, or other.
- You may use Loom (https://www.loom.com/)

For **sharing video**, you could use WeTransfer (https://wetransfer.com/).

Appendix B

Use a second Appendix to present any model(s)/ prototype(s) / results that were already submitted in the Interim Progress Demonstration (IPD) report. A summary of IPD models and findings could be included in the main body of the report.

Refer to the relevant Appendices in the main body of the report as appropriate (e.g. "For further details, refer to Appendix B").

Do not include the entire code in print as an appendix.